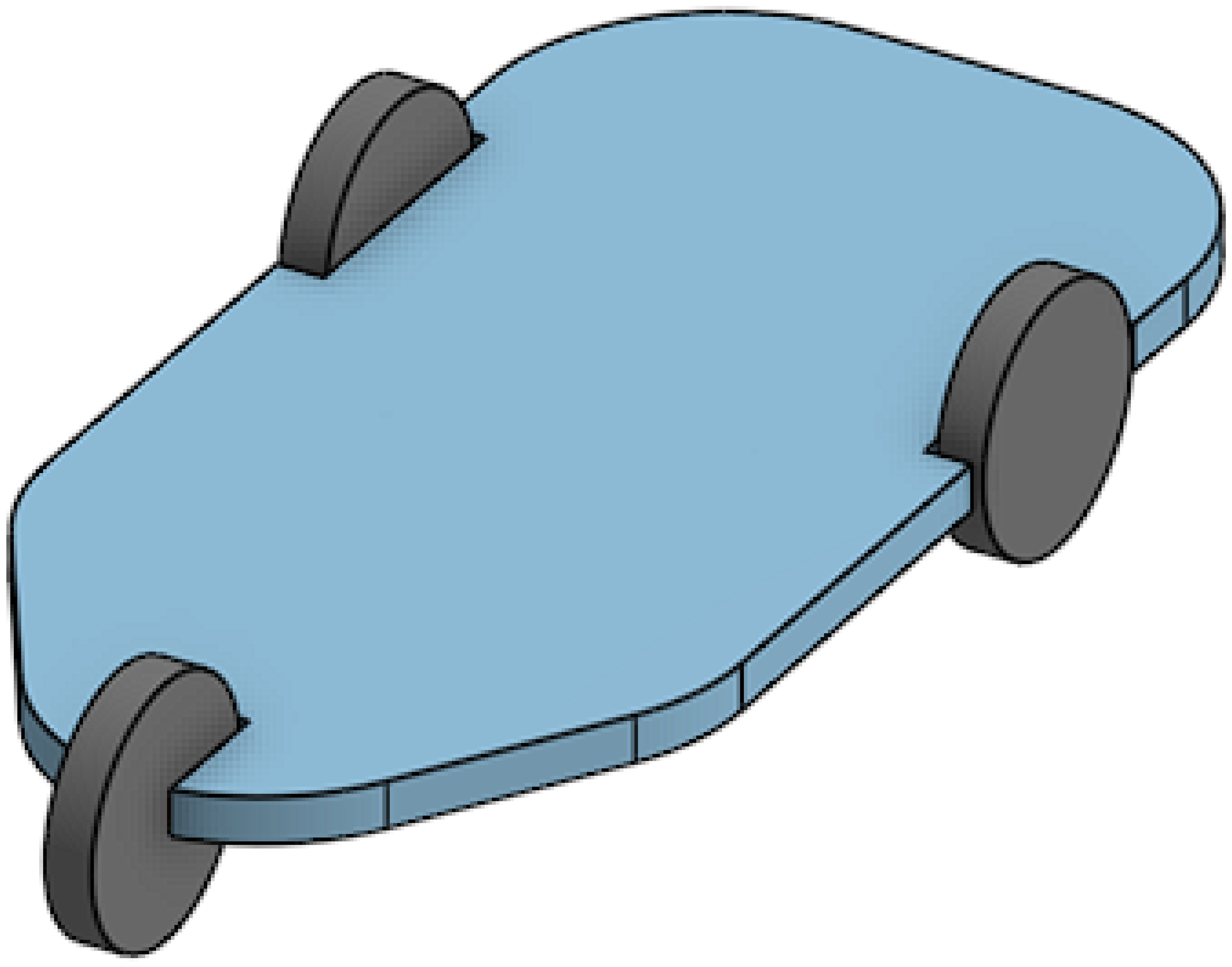
Problem Statement 2:

VEHICLE-LEVEL CONTROL CONCEPT FOR 3-WHEELER AUTO-RICKSHAW WITH INDEPENDENT REAR HUB MOTORS (DEEP-LEVEL EVALUATION)



1. **Introduction**

Electric three-wheelers are increasingly adopting hub motors due to their advantages in packaging flexibility, reduced mechanical complexity, and improved efficiency. However, removing the mechanical differential eliminates passive torque equalization between wheels.

As a result, asymmetric torque generation can introduce:

- Oversteer or understeer behavior
- Torque steer effects
- Yaw instability during transient maneuvers

Therefore, vehicle stability must be ensured electronically using a vehicle-level controller.

This work presents a control architecture capable of generating differential-like behavior using coordinated torque allocation between independent rear hub motors.

2. **System Overview**

The proposed system consists of a Vehicle Control Unit (VCU) that processes sensor inputs and generates torque commands for the left and right hub motors.

Sensor Used:

- IMU (for angle and acceleration measurements)
- Steering angle sensor (to measure Driver's turning angle)
- Encorder (for left and right motors to measure revolutions)

Control Outputs:

- Left motor torque Tl
- Right motor torque Tr

3. **Vehicle Dynamics Modeling**

The vehicle motion is described using longitudinal and yaw dynamics.
The longitudinal motion is governed by

$$m\dot{v} = \frac{T_L + T_R}{r_w} - F_{\text{drag}} - F_{\text{grade}}$$

where :

m = vehicle mass

rw = wheel radius

$$F_{\text{drag}} = \frac{1}{2}\rho C_d A v^2$$

$$F_{\text{grade}} = mg\sin(\theta)$$

where TL and TR are the torques applied by the left and right hub motors, respectively, and rw is the wheel radius.
Yaw motion arises due to torque imbalance between the wheels and is described by

$$Izr^{\cdot} = 2rwt(TR - TL)$$

where IzI_zIz is the yaw moment of inertia and ttt is the rear track width. This relation shows that any difference in wheel torque generates a rotational moment about the vehicle's vertical axis.

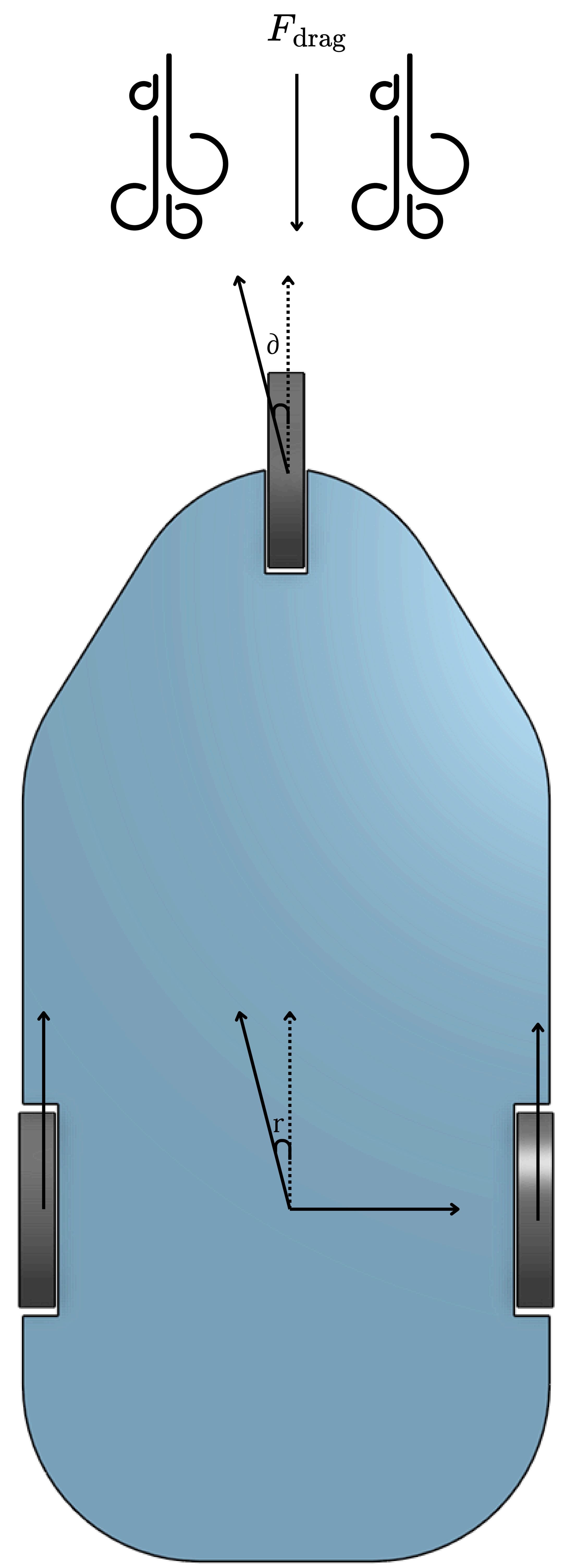


fig 1: vector diagram of force and angles

Yaw Stability Control
To ensure stable motion, the controller compares the desired yaw rate with the measured yaw rate:

$$e_r = r_{\text{ref}} - r$$

This yaw error indicates whether the vehicle is under-steering or over-steering relative to driver intent.
A feedback controller generates the desired corrective yaw moment:

$$M_{\text{des}} = I_z\left[K_p(r_{\text{ref}} - r) + K_d\frac{d}{dt}(r_{\text{ref}} - r)\right]$$

The controller therefore produces a stabilizing moment whenever the vehicle deviates from the intended trajectory.

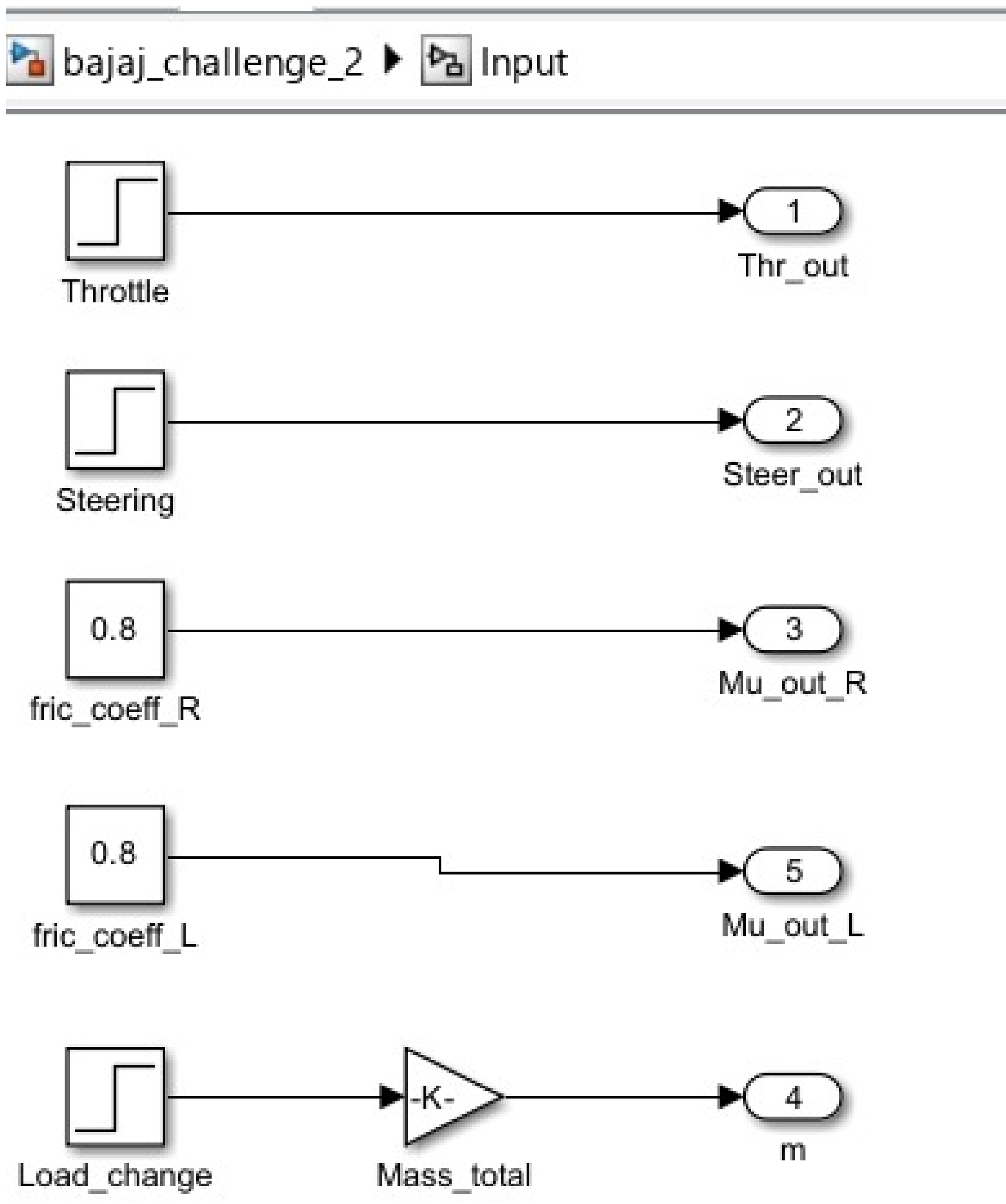


fig 2: Inputs to the system

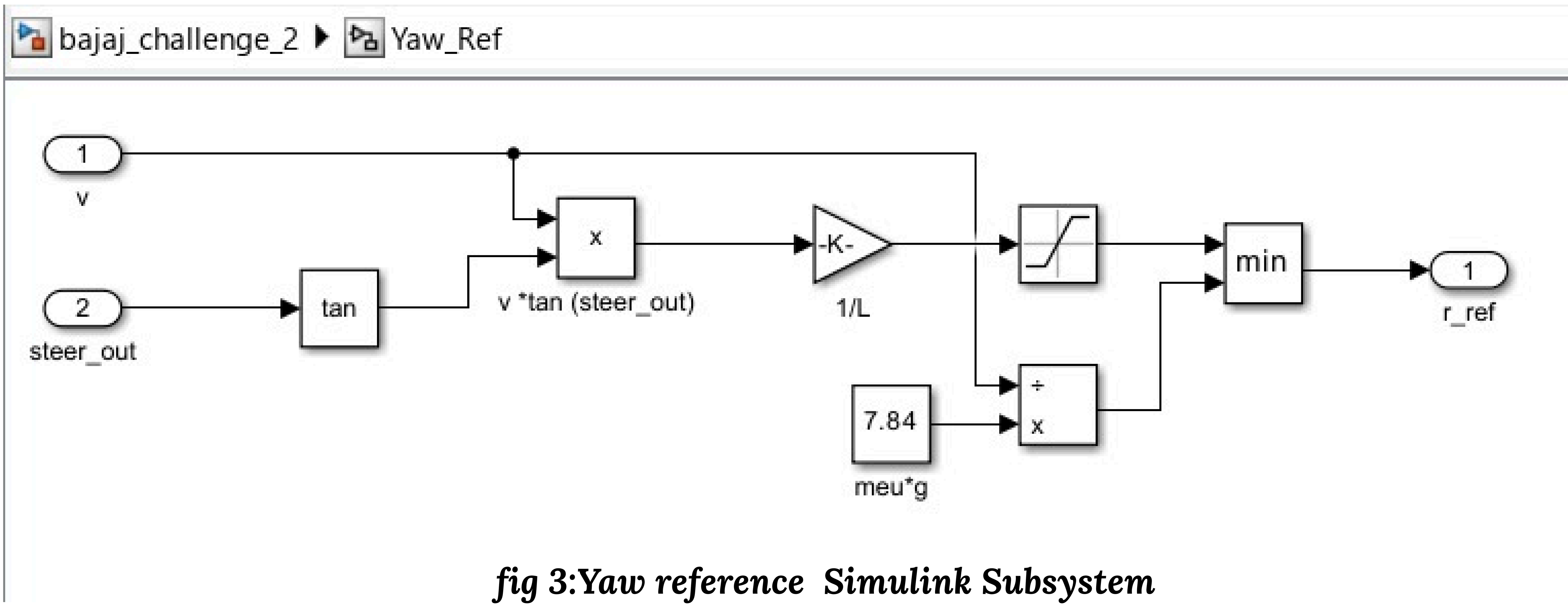


fig 3:Yaw reference Simulink Subsystem

Reference Yaw Model (Driver Intent)
We compute desired yaw rate from steering input:

$$r_{\text{ref}} = \frac{v}{L}\tan(\delta)$$

Where:
L = wheelbase
δ = steering angle

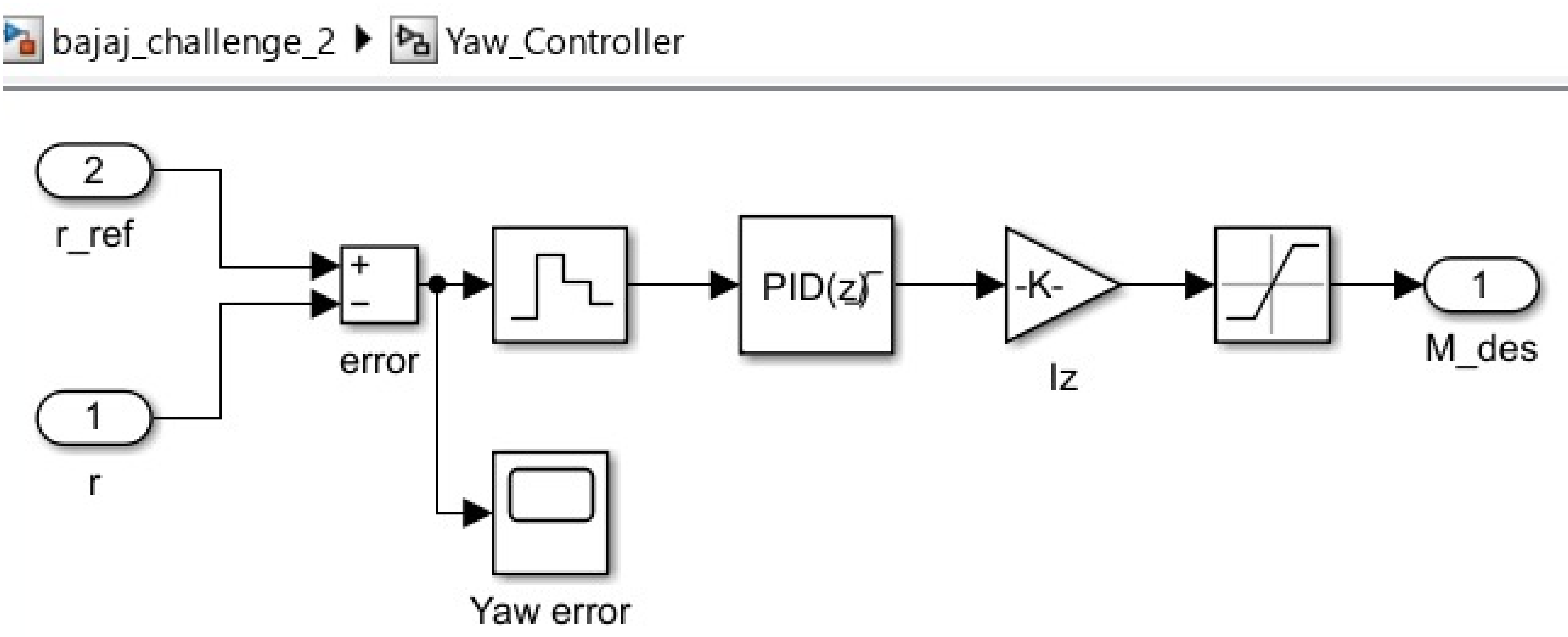


fig 4:Yaw controller Simulink Subsystem

Conversion of Yaw Moment to Torque Difference

The required yaw moment is achieved by creating a torque difference between the rear wheels. Using the yaw dynamics relation,

$$\Delta T = \frac{2r_w}{t} M_{\text{des}}$$

where ΔT\Delta TΔT represents the required torque differential between right and left wheels.

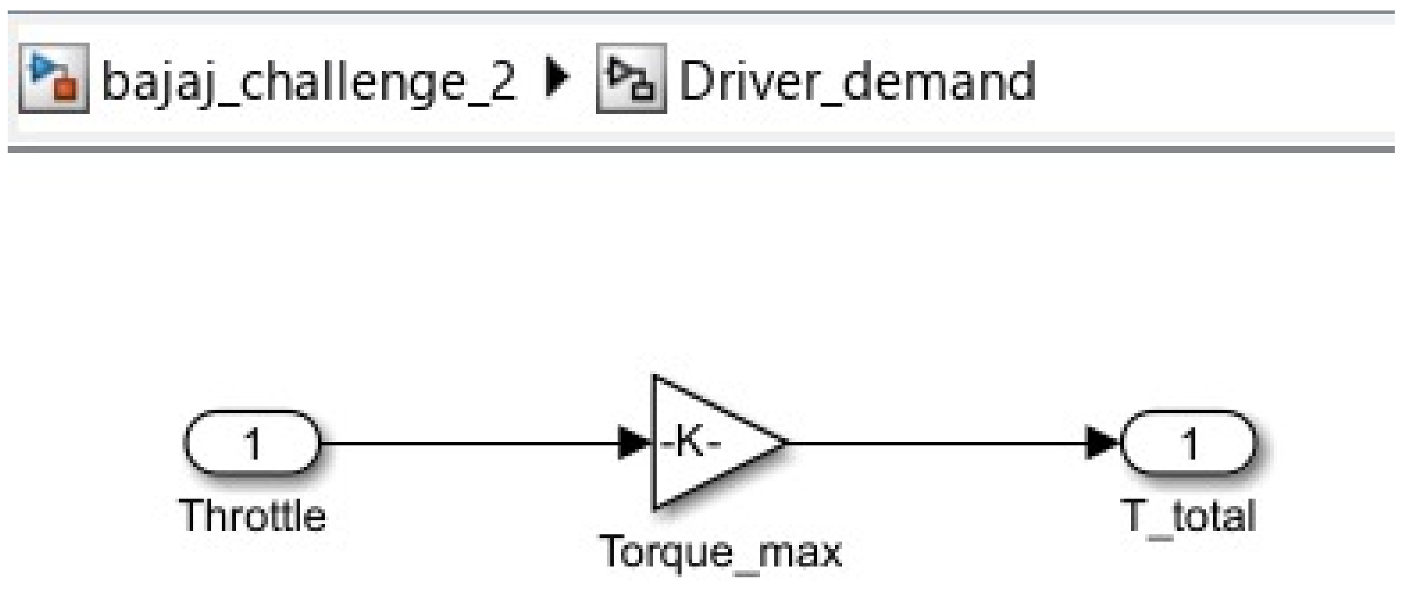


fig 5: throttle demand from driver

Driver Torque Demand

The throttle input determines the total propulsion torque required for longitudinal motion:

$$Ttot = f(throttle)$$

This torque represents the combined output of both hub motors needed to satisfy acceleration demand.

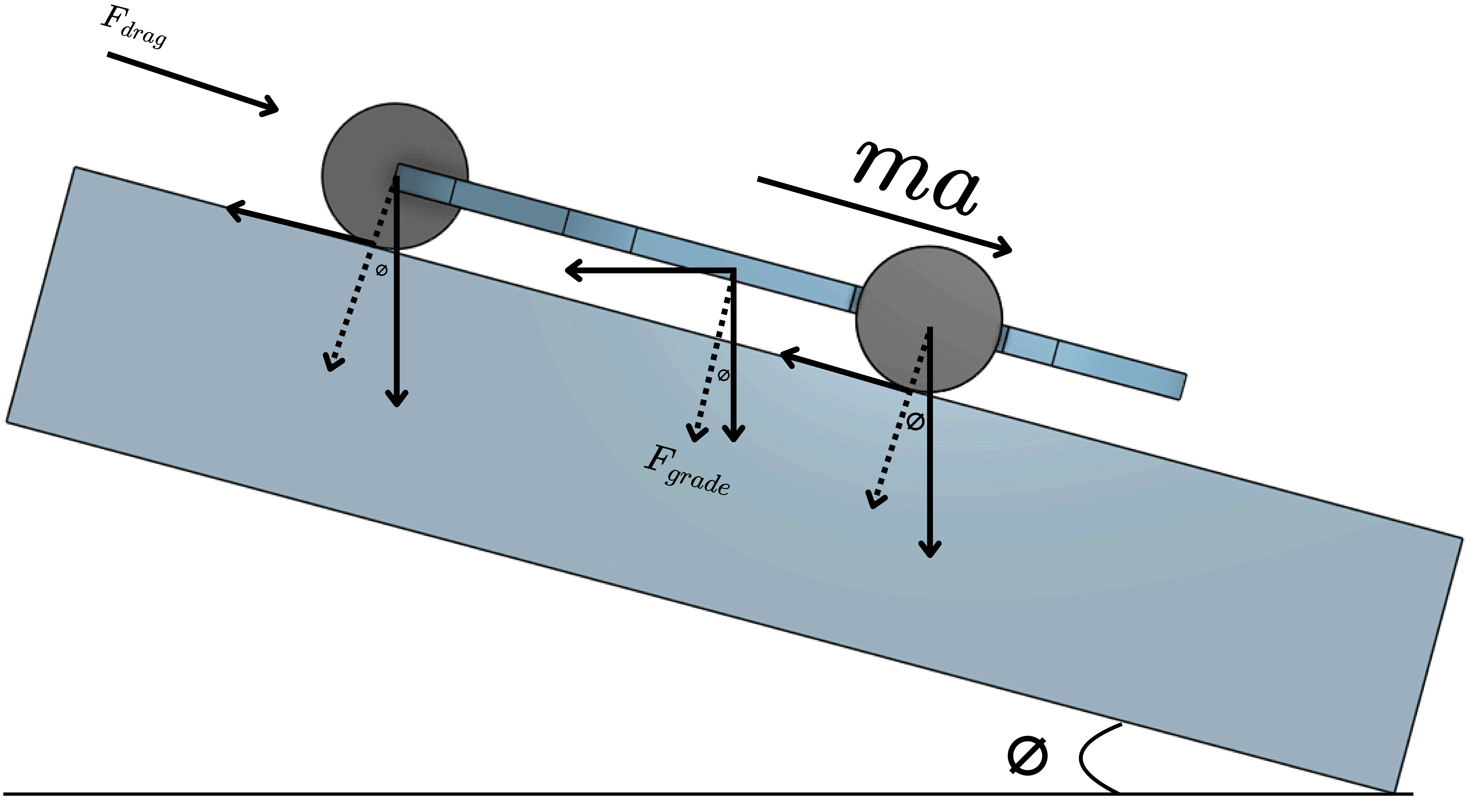


fig 6: Torque variation due to hill

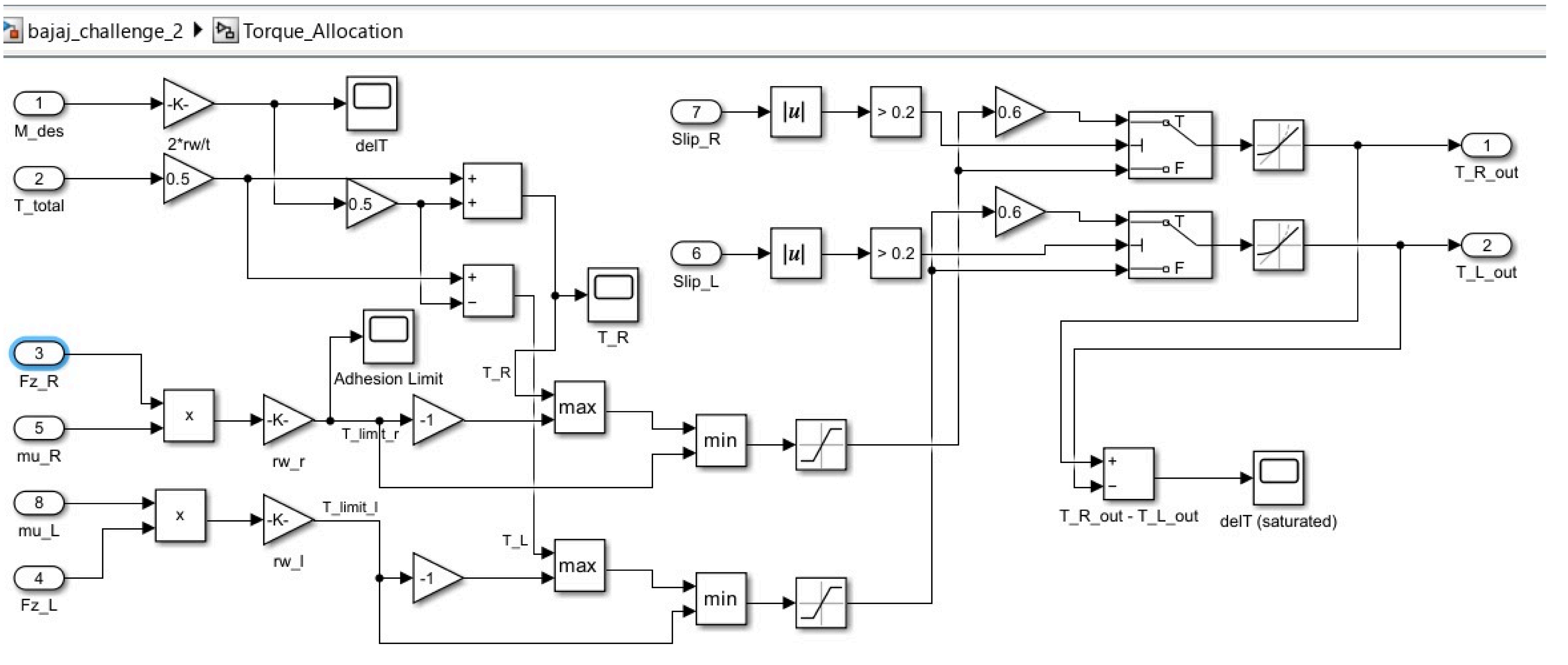


fig 7:torque allocation subsystem

Torque Allocation Strategy

The controller must simultaneously satisfy two conditions:

Total torque equals driver demand

$$T_R + T_L = T_{\text{tot}}$$

Torque difference produces required yaw moment

$$T_R - T_L = \Delta T$$

Solving these equations yields the torque allocation law:

$$T_R = \frac{1}{2}T_{\text{tot}} + \frac{1}{2}\Delta T$$

$$T_L = \frac{1}{2}T_{\text{tot}} - \frac{1}{2}\Delta T$$

Thus, wheel torque consists of a symmetric component responsible for forward motion and a differential component responsible for yaw control.

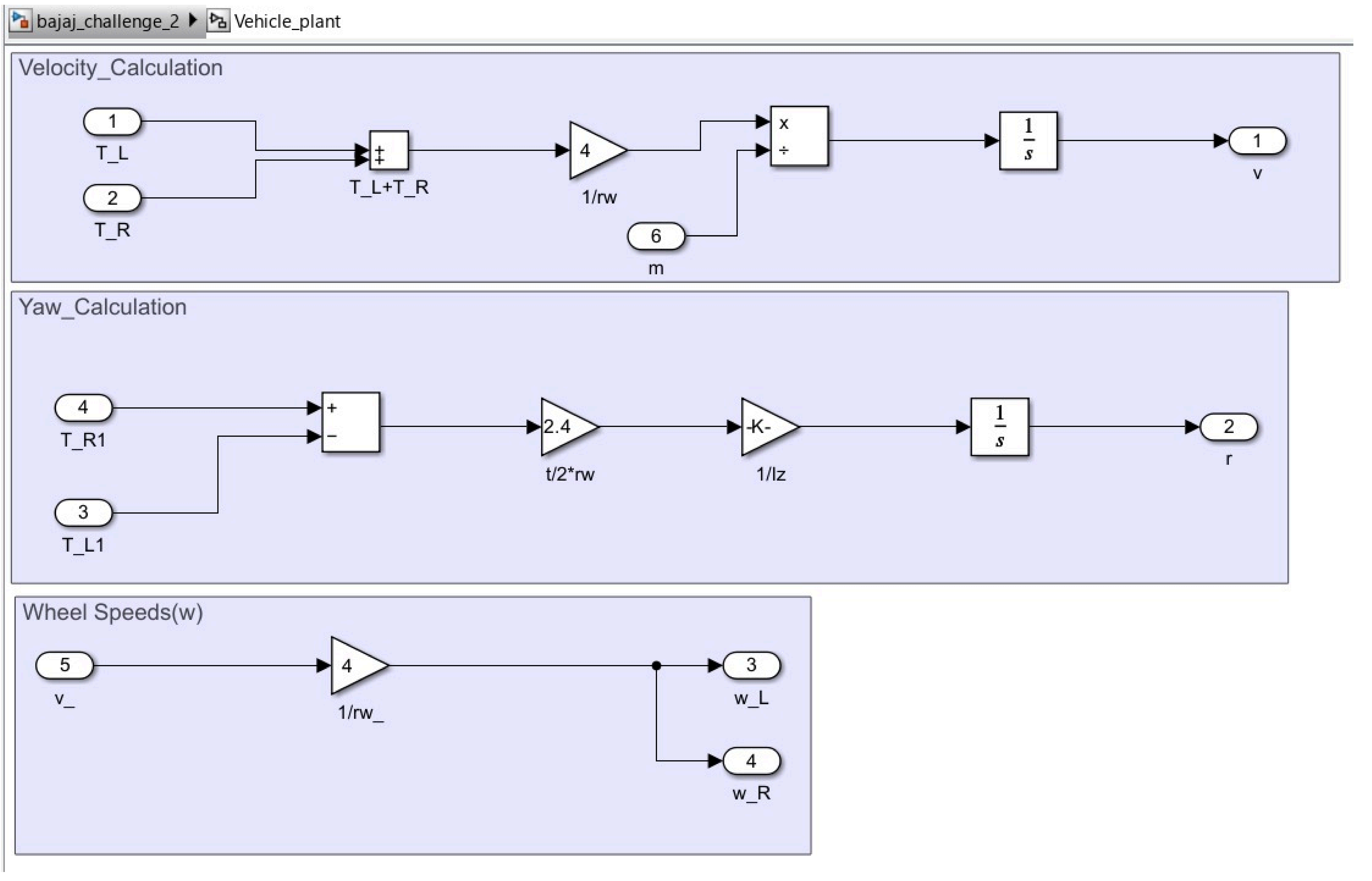


fig 8: Vehicle plant subsystem

Load Transfer and Traction Limitation

During cornering, lateral acceleration causes load transfer across the rear axle:

$$\Delta F_z = m_a y h_{cg}/t$$

The normal loads on the wheels become

$$Fz, R = 2mg + \Delta Fz$$

$$Fz, L = 2mg - \Delta Fz$$

The maximum allowable torque is limited by tire-road friction:

$$Tlimit = \mu F_z r_w$$

Torque commands are therefore saturated to prevent wheel slip and ensure stability under varying road conditions.

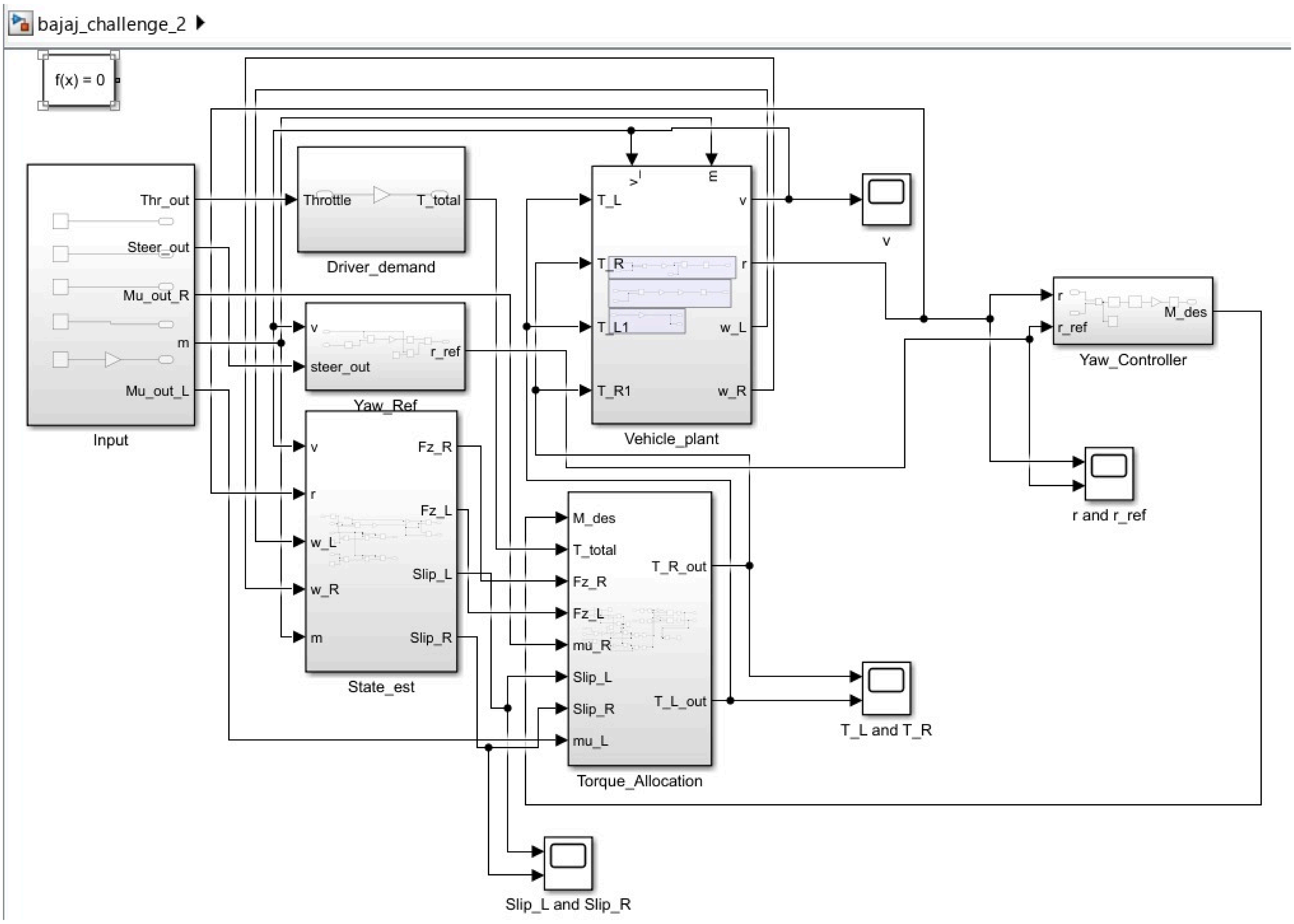


fig 9: overall system to control three wheeler

Oversteer / Understeer Logic

Oversteer

Condition: $r_{ref} > r_{ref} + margin$

Action:

Reduce ΔT

Possibly reduce Ttot

Understeer

Condition: $r_{ref} < r_{ref} - margin$

Action: $T(k) = T(k-1) + \mathrm{sat}\left(T_{\mathrm{cmd}} - T(k-1)\right)$

Increase ΔT(assist yaw)

Wheel Slip Estimation and Traction Control Logic
To ensure stable operation under varying road conditions and prevent excessive wheel spin, a slip-based traction monitoring mechanism is incorporated into the torque allocation framework. Wheel slip represents the difference between the wheel's circumferential speed and the actual vehicle longitudinal speed.
Slip Ratio Calculation
The slip ratio for each rear wheel is computed using wheel angular velocity measurements obtained from wheel speed sensors.
For the left wheel:

$$slip_L = (\omega_L r_w - v)/\ \mathrm{v}$$

For the right wheel:

$$slip_R = (\omega_R r_w - v)/\ \mathrm{v}$$

where:
ωL,ωR\omega_L, \omega_RωL,ωR = angular velocities of left and right wheels
rwr_wrw = wheel radius
vvv = vehicle longitudinal velocity

A positive slip value indicates that the wheel is rotating faster than the vehicle speed, implying loss of traction or wheel spin.
Slip Threshold Detection
To maintain tire-road adhesion, a slip threshold is defined. Excessive slip is detected when

$$\mid slip_i \mid > 0.2$$

where i∈{L,R}i \in \{L, R\}i∈{L,R}.

The threshold value of 0.2 corresponds to approximately 20% slip, which is commonly considered the upper bound of stable traction for typical road surfaces.
Torque Reduction Strategy
When excessive slip is detected, the controller reduces the torque applied to the slipping wheel to restore traction.
For the left wheel:

$$T_L = \begin{cases} 0.6\, T_L, & |\mathrm{slip}_L| > 0.2 \\ T_L, & \text{otherwise} \end{cases}$$

Similarly, for the right wheel:

$$T_R = \begin{cases} 0.6\, T_R, & |\mathrm{slip}_R| > 0.2 \\ T_R, & \text{otherwise} \end{cases}$$

This proportional torque reduction decreases wheel spin while maintaining forward propulsion

Overall Vehicle Control Architecture

Sensors
Collect real-time vehicle and driver input data required for control operation.
IMU
Measures vehicle motion parameters such as yaw rate and acceleration, used to monitor vehicle stability.
Wheel Speed Sensors
Measure rotational speed of each wheel to estimate vehicle speed and detect wheel slip.
Steering Angle Sensor
Measures steering input from the driver and indicates the intended turning direction.
State Estimation
Processes sensor data to estimate important vehicle states such as velocity, yaw rate, and slip with reduced noise and improved accuracy.
Driver Demand Generation
Converts throttle input into total propulsion torque required for vehicle acceleration.
Reference Yaw Model
Calculates the desired yaw rate based on steering angle and vehicle speed, representing expected vehicle motion.
Yaw Stability Controller
Compares desired and actual yaw rates and generates a corrective yaw moment to maintain directional stability.
Torque Allocation & Safety
Distributes total torque between left and right motors while applying safety constraints such as traction limits and slip control.
Left Motor Controller
Controls the left hub motor to produce the commanded torque accurately.
Right Motor Controller
Controls the right hub motor to produce the commanded torque accurately.
Vehicle Dynamics
Represents the physical vehicle response to applied torques, producing motion that is fed back to sensors.

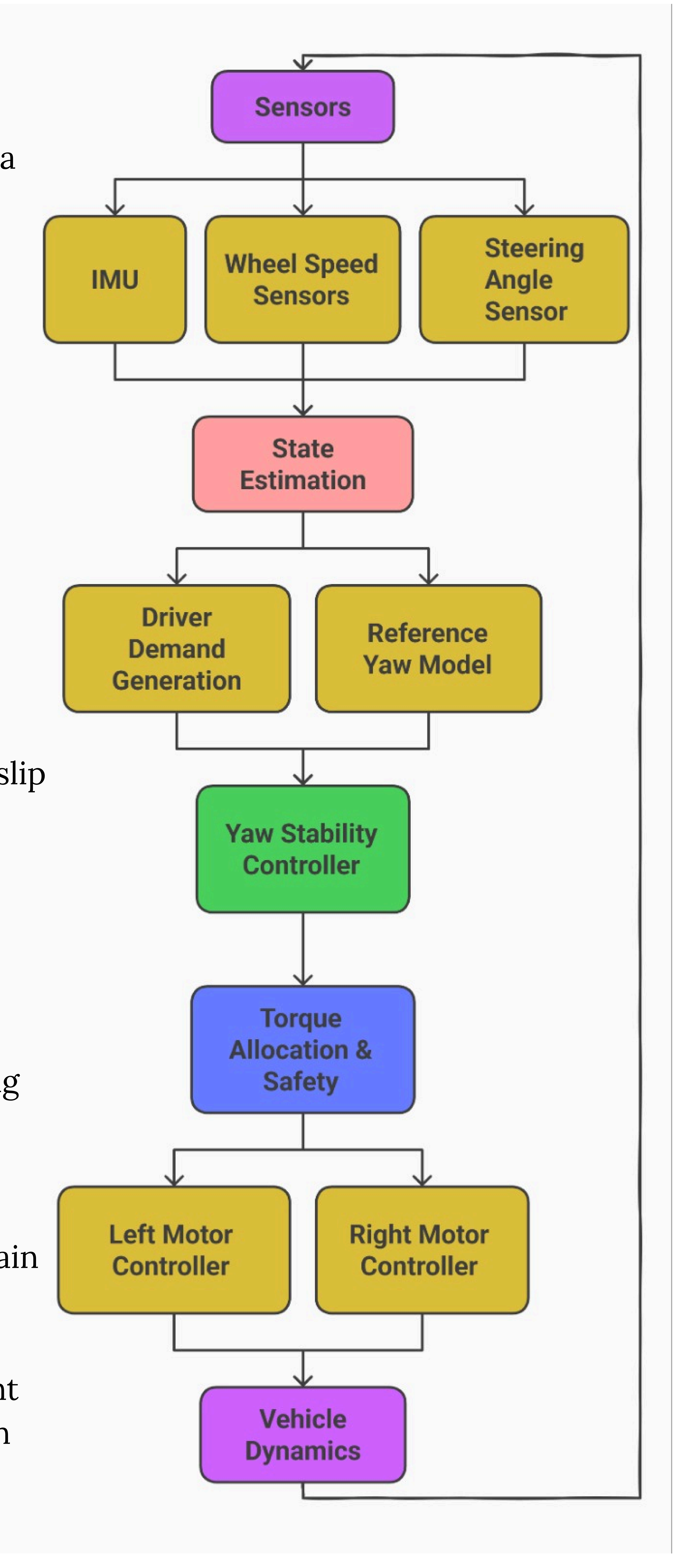


fig 10: overall system representation block diagram

Longitudinal Motion Case (Straight Driving)

In pure longitudinal motion, the steering angle is zero. Consequently,

$$r_{\text{ref}} = \frac{v}{L}\tan(\delta) = 0$$

and since the vehicle is not turning,

$$r = 0$$

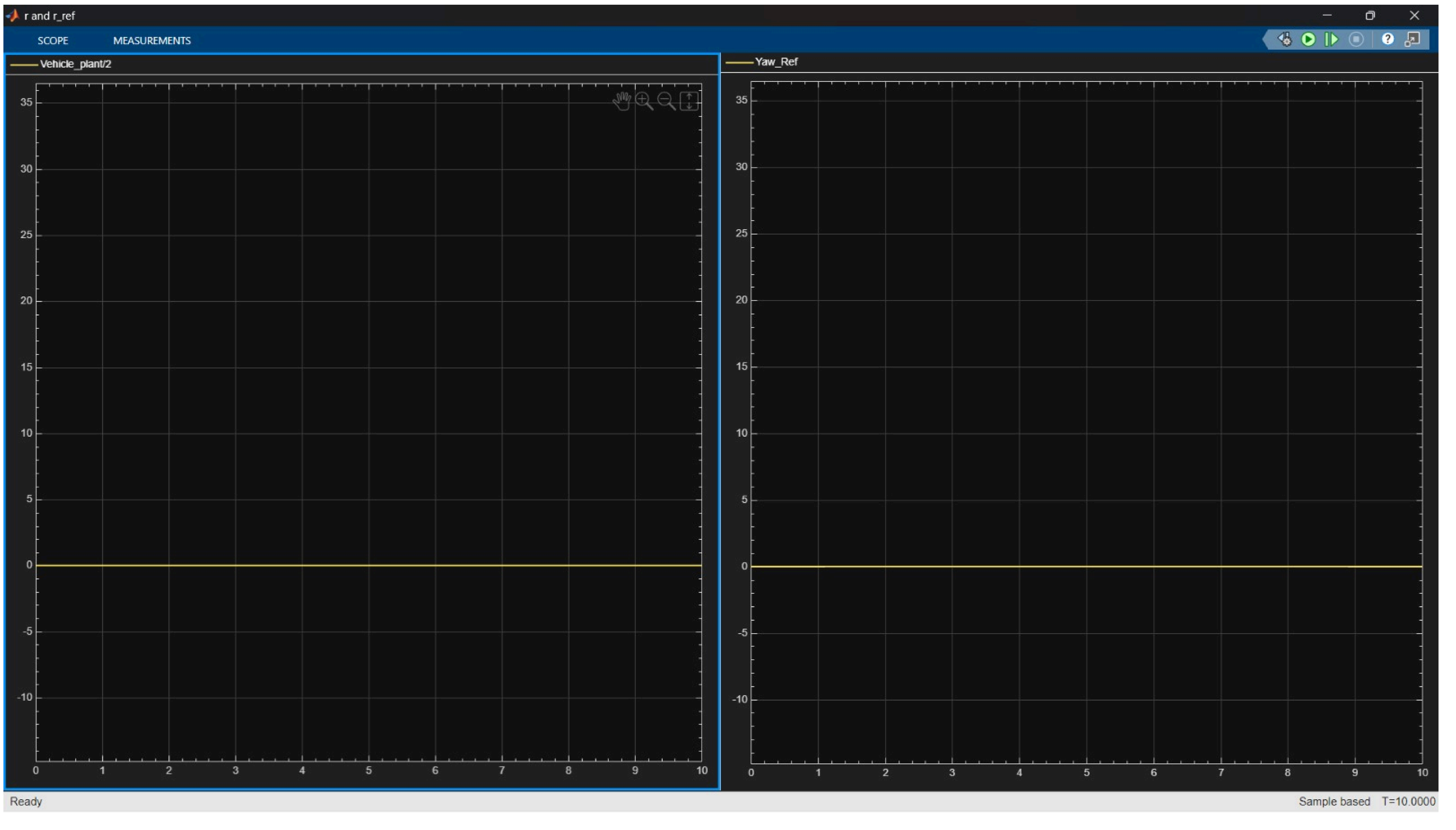


fig 11: graph of r and rref for longitudinal motion

The yaw error becomes zero, producing no corrective yaw moment:

$$M_{\text{des}} = 0$$

This leads to zero torque differential:

$$\Delta T = 0$$

Substituting into the torque allocation equations gives

$$T_R = T_L = \frac{1}{2}T_{\text{tot}}$$

Hence, both wheels receive equal torque, generating only forward propulsion without rotational motion. Under this condition, the torque vectoring controller behaves equivalently to a conventional differential.

Turning Motion Case (Cornering Driving)

During turning motion, the driver applies a non-zero steering angle, causing the vehicle to follow a curved trajectory. The steering input generates a desired yaw rate based on vehicle kinematics, given by

$$r_{\text{ref}} = \frac{v}{L}\tan(\delta)$$

The actual vehicle yaw rate may differ from the desired value due to disturbances, load variations, or road conditions. Therefore, the yaw error is calculated a

$$\delta \neq 0 \quad \Rightarrow \quad r_{\text{ref}} \neq 0$$
$$e_r = r_{\text{ref}} - r$$

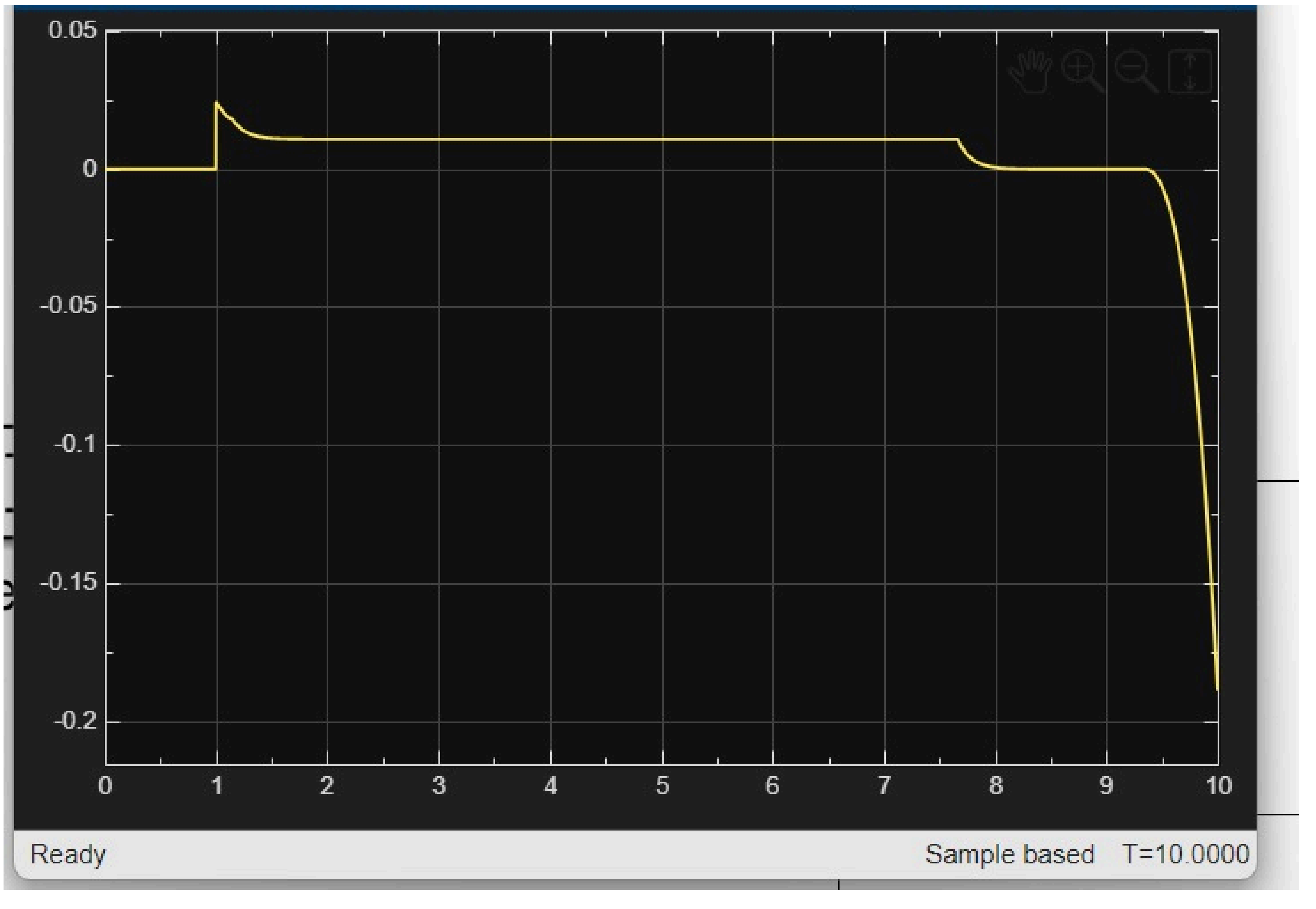


fig 12:Yaw error data during motion

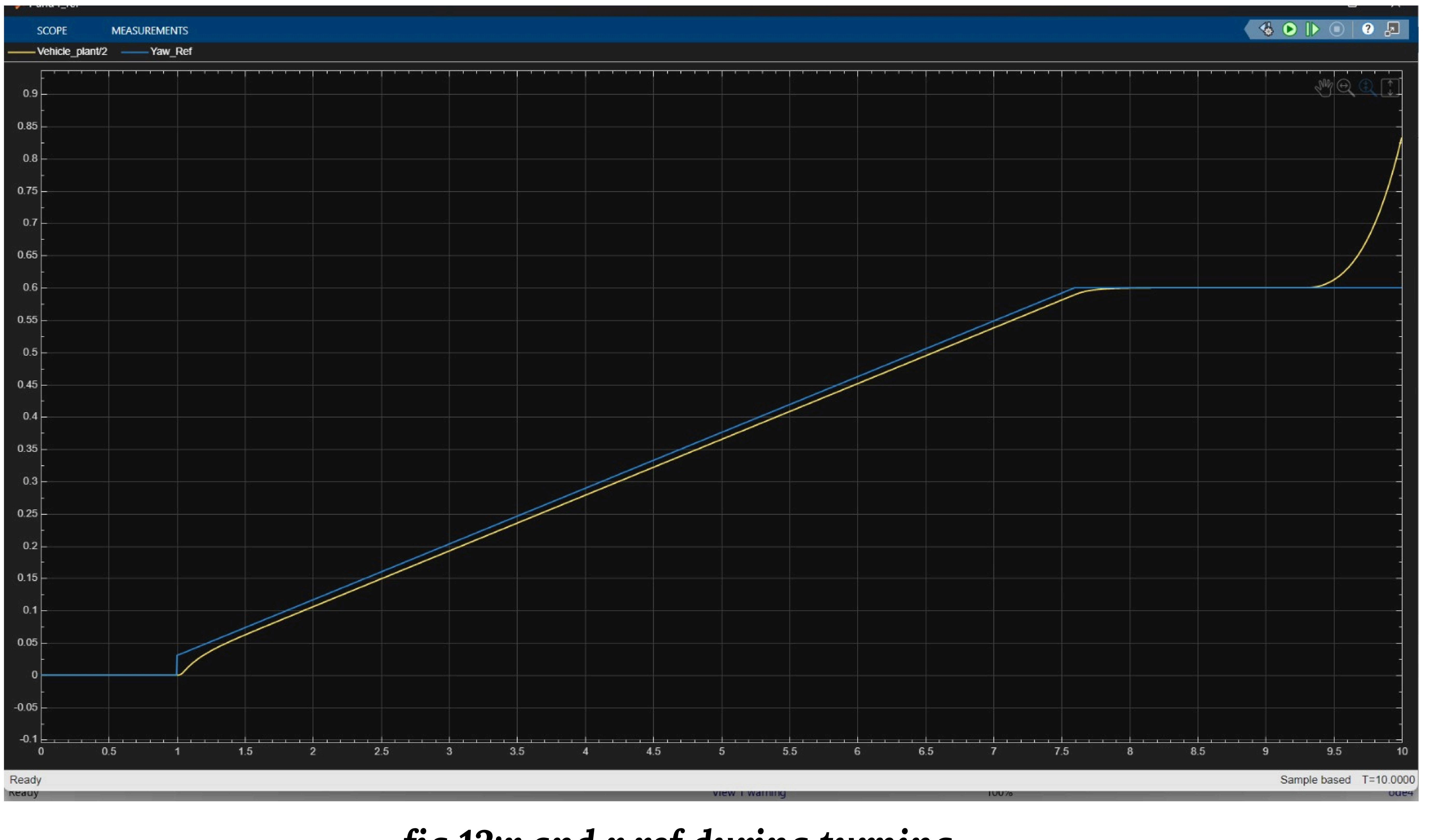


fig 12:r and r ref during turning

This error is processed by the yaw stability controller to generate the corrective yaw moment required to align the vehicle motion with driver intent:

$$M_{\text{des}} = I_z \left(K_p e_r + K_d \dot{e}_r \right)$$

The desired yaw moment is achieved by creating a torque difference between the left and right rear hub motors. The required torque differential is obtained from vehicle yaw dynamics as:

$$\Delta T = \frac{2r_w}{t} M_{\text{des}}$$

Because a turning motion is required,

$$\Delta T \neq 0$$

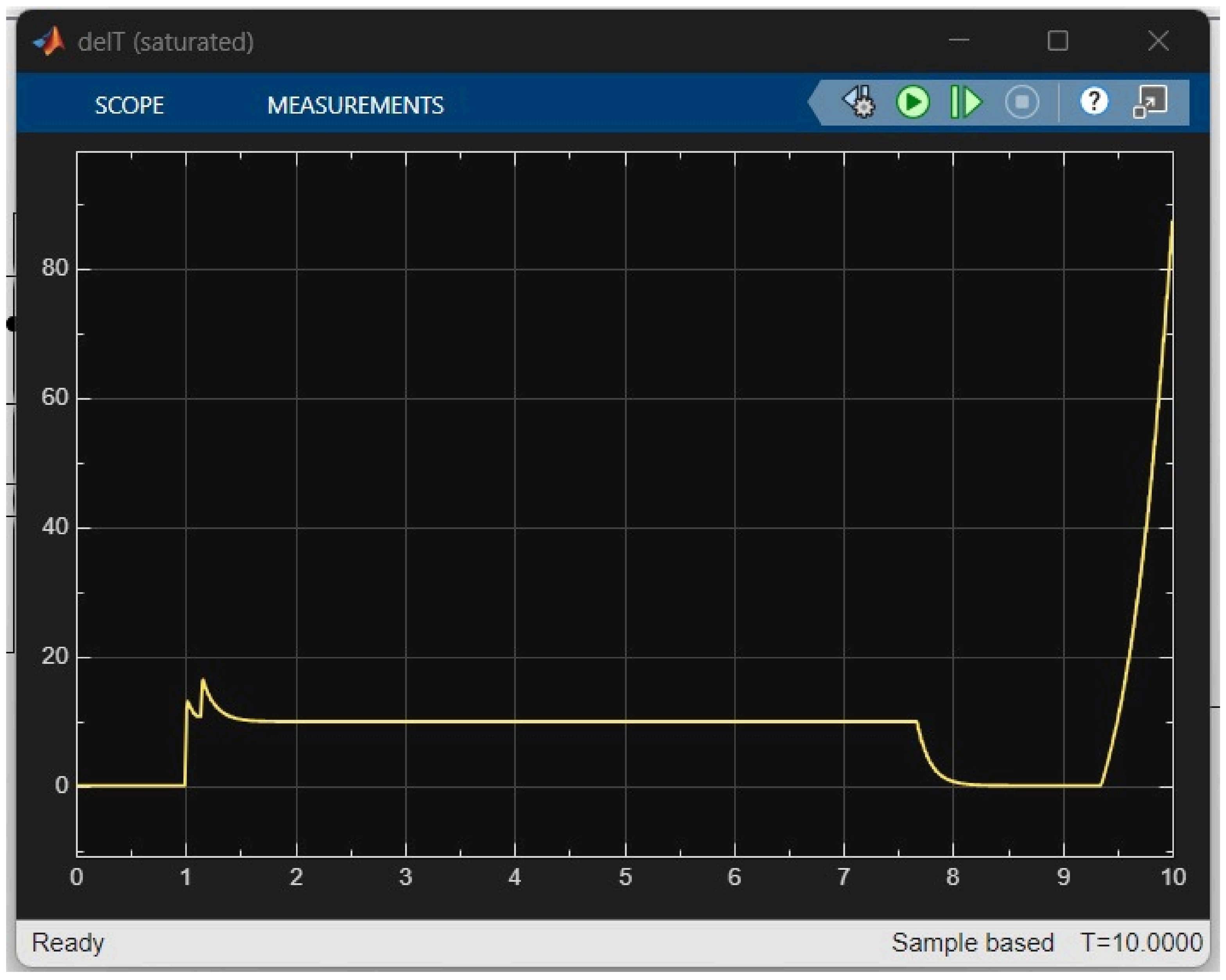


fig 13: Difference between torque left and torque right

The total propulsion torque demanded by the driver is then distributed between the two wheels according to the torque allocation law:

$$T_R \neq T_L$$

$$T_R = \frac{1}{2} T_{\text{tot}} + \frac{1}{2} \Delta T$$

$$T_L = \frac{1}{2} T_{\text{tot}} - \frac{1}{2} \Delta T$$

which produces a yaw moment about the vehicle's center of gravity:

$$I_z \dot{r} = \frac{t}{2r_w} (T_R - T_L)$$

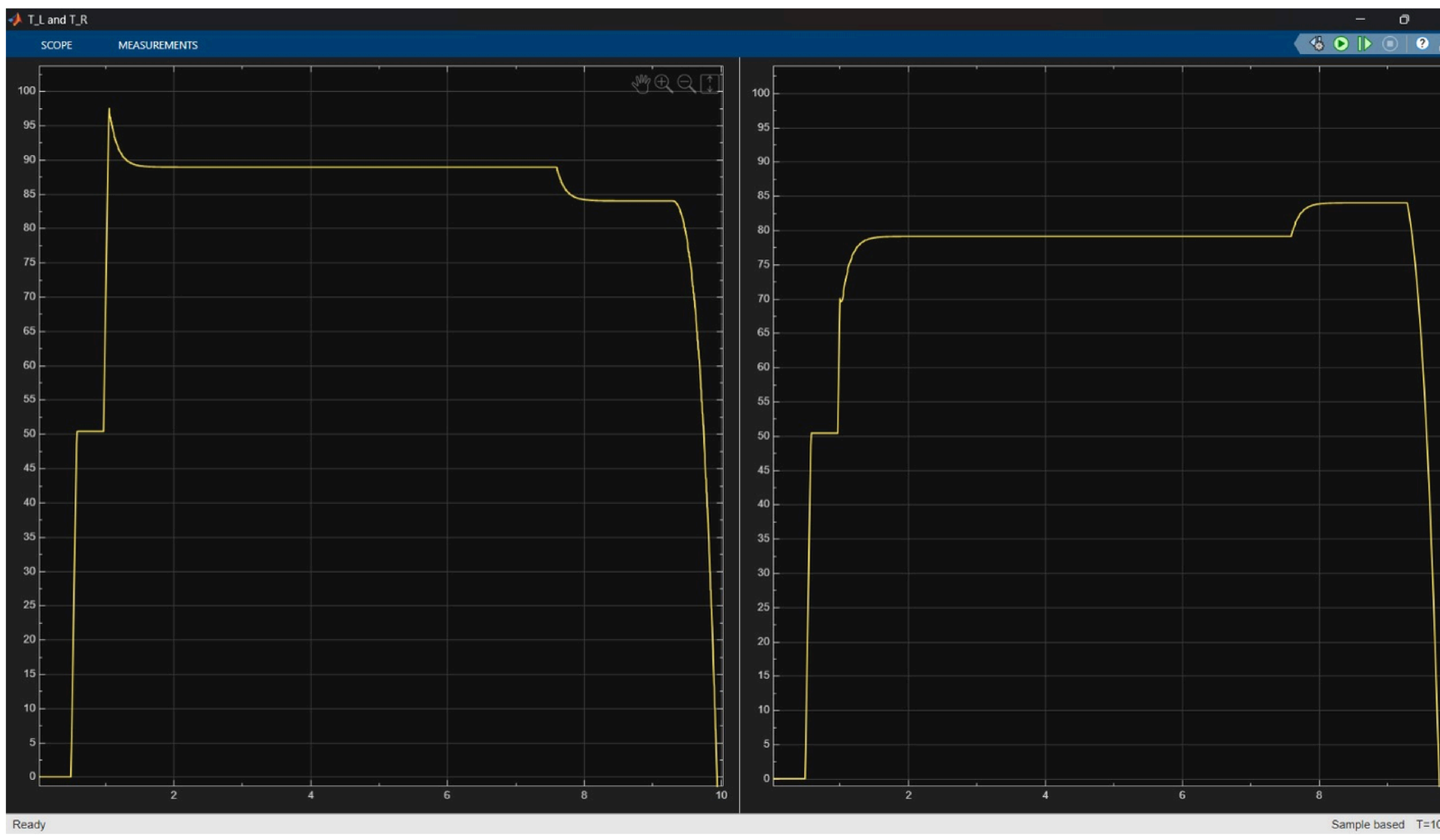


fig 14: Graph of torque left and torque right

This differential torque enables controlled rotation of the vehicle, allowing it to accurately follow the intended turning path. Therefore, during cornering conditions, the torque vectoring controller actively redistributes torque between the wheels to generate the required yaw motion while maintaining vehicle stability and driver steering expectations.

Thank You

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